

SKILLSIGNALZA

Data Analytics Career Map Pack

A South African entry-level route guide

Choose a lane. Build working analysis. Publish credible proof. Apply with a system.

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START HERE

What this pack will help you finish

Data Analytics becomes easier to navigate once you separate job labels from the work behind them. This pack gives you a route into that work. You will choose a lane, set a tool boundary, build evidence that can be inspected, and use job adverts to guide each improvement cycle.

By the final page, you will have recorded:

- The analytical work you want to target first.
- The job titles and search terms linked to that work.
- The SQL, spreadsheet and BI skills required for your current lane.
- The secondary tools that deserve time during this build cycle.
- The project that will carry your strongest evidence.
- The acceptance checks that determine when the project is complete.
- The application routine you will run each week.
- The date when you will review results and adjust the route.

WORKING PROMISE: Complete the decisions and worksheets. You will finish with a focused route, a defined evidence build, and a scheduled application cycle.

Product boundary

Included in this edition:

- South African entry-level Data Analytics market direction.
- Lane selection based on analytical work and deliverables.
- A practical priority order for SQL, Excel, Power BI and supporting tools.
- Project blueprints with business questions, outputs and completion checks.
- Evidence packaging, application tracking and a 30/60/90-day route.

Reserved for the SkillSignalZA Readiness Report:

- An individual readiness score or score band.
- Personal CV assessment or rewriting.
- Inspection of a buyer's portfolio, GitHub, Kaggle or dashboard links.
- Candidate-specific gap diagnosis and project selection.
- Recruitment, vacancy matching, salary forecasting or hiring predictions.

NAVIGATION

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Recommended reading order

Complete Starting Point before choosing a lane. Read the full Market Evidence section. Compare the lane outputs, commit to one direction for the current evidence cycle, and keep that choice visible while working through the project and application sections.

KEEP ONE WORKING DIRECTION: Change lanes after a completed project and an application review. Repeated switching breaks the connection between learning, evidence and target roles.

HOW TO USE THE PACK

Keep an editable document or notebook open while reading. Every page should change a decision, a task, an artifact or an application action.

- 1 Record the evidence you already possess.
- 2 Select the route that matches your present level of control.
- 3 Choose a lane and define the tool boundary for this cycle.
- 4 Build or repair one project against a visible completion gate.
- 5 Package the same evidence consistently across your CV and submitted links.
- 6 Run targeted applications and review recurring gaps each week.

The evidence ladder

Judge each skill using four levels:

Level	Evidence standard	Direct check
Named	The skill appears in your CV, course record or notes.	Point to the exact claim.
Explained	You can describe its purpose, limits and place in a workflow.	Explain it in plain language.
Applied	You have used it to complete a defined analytical task.	Open the query, workbook, report or script.
Demonstrated	Another person can inspect the work, follow the process and understand the result.	Share an accessible artifact with context.

Named skills help keyword matching. Applied and demonstrated evidence carries the application because it proves you can move from a question to a defensible result.

Course rule

Start a course when it fills a named gap in your chosen lane. Connect each learning block to a working artifact. A completed course with no query, workbook, dashboard, analysis note or project output leaves the application evidence unchanged.

STARTING POINT

Record what already exists

Use exact evidence. Write the file, link, query, dashboard or work example beside every claim.

Area	Current level	Evidence you can open	Main gap
SQL			
Spreadsheet analysis			
BI and visualisation			
Interpretation and statistics			
Cleaning and preparation			
Reporting and communication			

Run a four-part foundation check

For each area above, test whether you can:

- Explain the purpose and the decision it supports.
- Complete a small task without following a copied sequence line by line.
- Check the result for accuracy, missing values and misleading interpretation.
- Show the output with enough context for another person to follow it.

Write the first point where your control breaks:

Write the strongest piece of analytical evidence you already have:

Write the skill you currently claim with the weakest supporting context:

STARTING POINT

Choose your entry route

Route	Choose this route when	First objective
Foundation Builder	You struggle to query, analyse, validate or explain a small dataset independently.	Build control across one complete analytical workflow.
Evidence Builder	You can complete the work, while your public or submitted proof remains thin, scattered or unclear.	Repair the shortest path to credible visible evidence.

Candidates with strong visible work can enter at Package and Apply. They should still complete the lane decision and project-completion check before sending applications.

A direct placement test

Use one dataset with at least two related tables or one flat file with enough columns to answer a business question. Complete these checks without a step-by-step tutorial:

- 1 Write SQL or an equivalent query process that filters, joins or groups the data correctly.
- 2 Clean a documented issue and preserve the raw source.
- 3 Build a useful summary in Excel or a BI tool.
- 4 State the finding, the limitation and the action a stakeholder could take.

Choose Foundation Builder when the workflow breaks before the final result. Choose Evidence Builder when the work is sound and the proof needs packaging.

ROUTE DECISION: Select the route that matches today's evidence. The label can change after a completed milestone.

FOUNDATION BUILDER

Establish independent control

Your first goal is a repeatable path from raw data to an explained result. Tool collection can wait. Build one workflow that you can perform, check and describe.

Use this order:

- 1 Read the business question and define the required output.
- 2 Inspect the data structure, field meanings and obvious quality risks.
- 3 Query or filter the records needed for the analysis.
- 4 Clean and transform the working copy while preserving the raw source.
- 5 Calculate the relevant measures and test them against known totals.
- 6 Build a report or visual that serves the question.
- 7 Write the finding, limitation and recommended next action.
- 8 Save the files so another person can reproduce the result.

Foundation build rule

Stay with one primary stack for the current cycle:

- SQL for extraction and structured analysis.
- Excel for inspection, calculations and quick summaries.
- Power BI for modelling, measures and dashboard communication.
- A named database environment for realistic query practice.

Add Python or R when your selected lane or project creates a clear reason. The core workflow already contains enough depth for a strong first evidence build.

Advance when

You can start from a fresh copy of the data, complete the workflow without copied instructions, explain every major step, and identify at least one limitation in your own result.

EVIDENCE BUILDER

Repair the shortest path to credibility

Begin with the best work you already have. A repair cycle should upgrade evidence faster than restarting an entire learning path.

Audit one project against these questions:

- Can a reviewer identify the business question within one minute?
- Is the dataset source or synthetic-data declaration visible?
- Are the raw data and transformed data kept separate?
- Can the reviewer inspect the SQL, spreadsheet logic, transformation steps or script?
- Are calculations and dashboard measures defined?
- Do the findings answer the stated question?
- Are limitations and assumptions written down?
- Can another person open the work through the submitted link?

Repair order

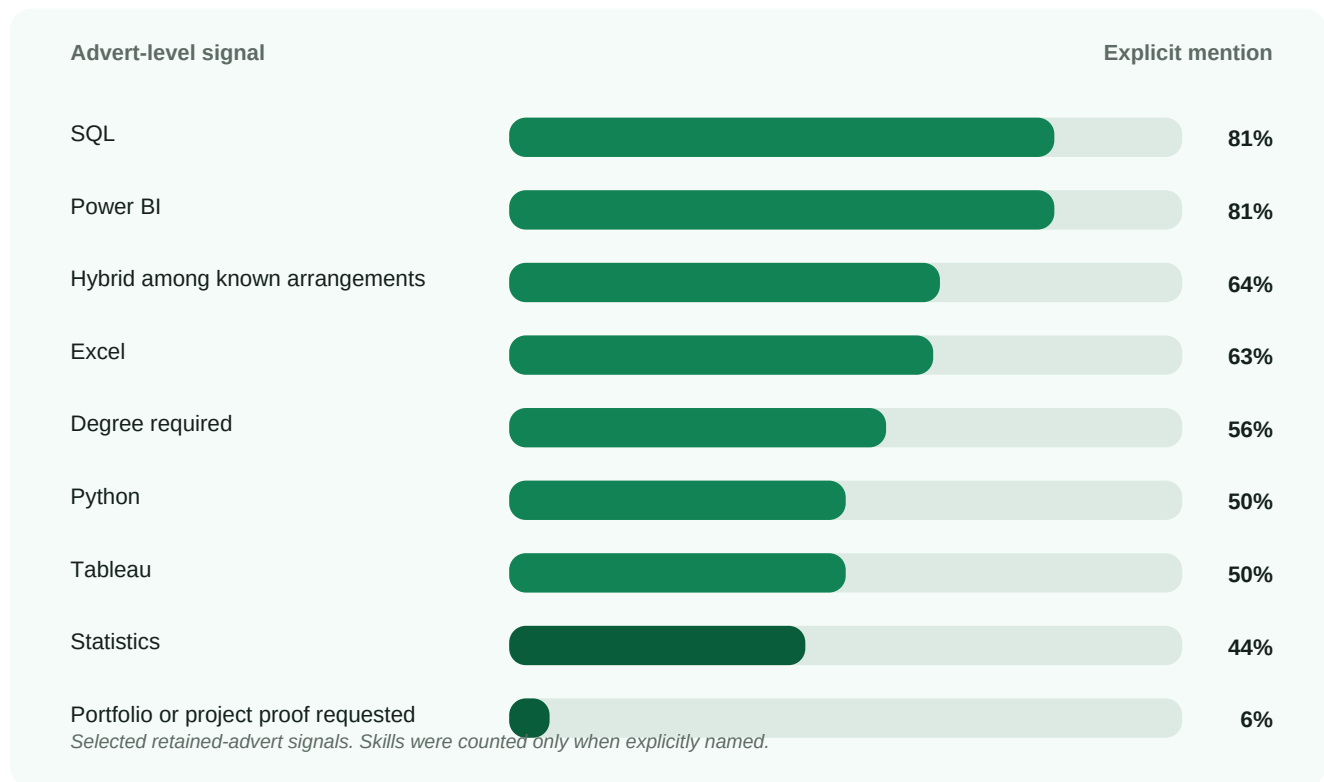
Fix the highest broken layer first:

Layer	Common failure	Required repair
Access	Private, expired or incomplete link	Publish an accessible project location and test it in a signed-out browser.
Context	Dashboard appears with no question or dataset explanation	Add a short project brief, source note and intended user.
Process	Final chart exists while the analysis path is hidden	Include queries, cleaning notes, transformations and validation checks.
Interpretation	Visuals repeat numbers without meaning	Write findings, limits and a decision or action supported by the data.
Reproduction	Files depend on an unexplained local setup	Add file order, refresh steps and required software.

EVIDENCE BUILDER RULE: Repair one project to completion before opening a second repair candidate.

MARKET EVIDENCE

What appeared in the retained Data Analytics adverts



The percentages show explicit advert-level mentions in the retained South African entry-level-accessible sample. They describe the evidence used for this pack. They carry no claim of market-wide representation.

Direct readings

- SQL and Power BI form the strongest repeated technical combination.
- Excel remains part of the working core, especially for dashboards, pivots and recurring reports.
- Python and Tableau strengthen several routes and sit below the first-build priority line.
- Statistics supports interpretation and analytical investigation.
- Portfolio requests appeared rarely, while visible work still improves the quality of a candidate's application evidence.

MARKET EVIDENCE

Turn frequency into a build order

Frequency helps you decide what to build first. It never replaces lane choice or job-specific reading.

Priority	Build target	Why it sits here	Evidence output
First	SQL	It was the most consistent core analytical skill and appeared across several role labels.	Query file showing filters, joins, grouping and validation.
First	Power BI or comparable BI capability	Dashboard and reporting work appeared repeatedly, with Power BI leading captured tools.	Interactive report, exported view and metric definitions.
First	Excel analysis	Spreadsheet capability remained common and often connected to reporting work.	Auditable workbook with formulas, pivots or Power Query steps.
Next	Analysis, interpretation and statistics	Tools produce outputs; analytical judgment turns them into findings.	Written insight note with assumptions and limits.
Next	Data cleaning and preparation	Every credible project needs traceable handling of messy data.	Cleaning log, transformed dataset and quality checks.
Lane-based	Python or R	Analytical programming supports deeper investigation, automation and larger workflows.	Script or notebook with readable steps and outputs.
Lane-based	Tableau, Looker Studio or related tools	Comparable BI tools expand tool flexibility after the reporting workflow is secure.	Second report or a deliberate rebuild of the same analysis.

The default route for an undecided beginner

Choose SQL + Excel + Power BI for the first complete evidence cycle. Use one dataset across the stack. Extract with SQL, inspect and validate in Excel, model and report in Power BI, then publish the files with a written interpretation.

That route aligns with the strongest recurring signals and produces evidence for more than one entry-level title.

MARKET EVIDENCE

Qualifications, experience and location

Formal qualifications influence access to advertised roles. In the retained sample, degree requirements appeared in just over half of the adverts. Other adverts left qualifications unspecified, marked a degree as preferred, or mentioned a certificate or bootcamp route.

Technical evidence has a separate job: showing that you can complete analytical work. Use both parts of the application deliberately.

Your position	Application treatment
Relevant qualification completed	State the qualification, institution and completion year clearly. Connect coursework to specific analysis evidence where useful.
Relevant qualification in progress	Include the expected completion date, completed modules with direct relevance, and current project evidence.
Bootcamp or certificate route	Name the programme and pair it with accessible work showing SQL, spreadsheet and BI use.
Equivalent experience	Describe the analytical responsibility, tools used, decision supported and output produced.
Limited formal access	Prioritise adverts with flexible qualification wording and build stronger direct evidence around the working core.

Experience wording

The retained adverts included internships, graduate entry, junior roles and associate titles. Experience wording ranged from zero years through three years, while several adverts left the requirement unspecified.

Apply when you can perform a meaningful share of the work and meet the central technical signals. Treat a short experience range as one factor in the advert. Use the CV to make internships, coursework, employment tasks and self-directed projects legible as evidence.

Geography and work arrangements

Gauteng held the largest share of retained locations. Western Cape roles also appeared, with additional evidence from Eastern Cape. Hybrid work led among adverts with a known arrangement. On-site roles formed a meaningful secondary group, while remote roles were scarce.

Build the search around places you can actually work. Include nearby city names and hybrid or on-site filters. Keep remote applications in the mix when the role fits, while planning the weekly search around the larger location pools.

APPLICATION RULE: Read the full advert. Tool combinations, qualification wording, experience ranges and work arrangements vary by role.

SHARED FOUNDATIONS

Build SQL deeply enough to investigate data

SQL sits at the centre of the DA route. Your evidence should show control over questions and results, beyond a list of commands.

Working SQL scope

Build these capabilities in order:

- 1 Select named fields and filter rows with conditions you can explain.
- 2 Sort results, handle nulls and create clear calculated fields.
- 3 Aggregate with `COUNT`, `SUM`, `AVG`, `MIN` and `MAX` at the correct grain.
- 4 Group records and use `HAVING` when the filter belongs after aggregation.
- 5 Join related tables while checking key uniqueness and row-count changes.
- 6 Use subqueries or common table expressions to separate reasoning steps.
- 7 Apply window functions for ranking, running totals or comparisons when the question requires them.
- 8 Validate outputs against source totals and documented expectations.

Query discipline

Record the grain of every table before joining. A customer table may contain one row per customer while an order table contains many rows per customer. Joining them changes row counts by design. State that change and test it.

Use readable names, consistent indentation and comments for business rules. Save the final queries in execution order. Keep exploratory queries when they explain an important decision or failed assumption.

SQL evidence gate

Your project should contain:

- A named business question.
- A short schema or field guide.
- Queries using filters, aggregation and at least one defensible join.
- Validation checks for row counts, duplicates and nulls.
- A final dataset or result table consumed by the report.
- A note explaining one query decision and its effect on interpretation.

SQL COMPLETION CHECK: You can explain the source grain, join logic, measure definition and validation result without opening a tutorial.

SHARED FOUNDATIONS

Use spreadsheets as an analytical workspace

Excel appears in the market because analysts still receive extracts, recurring files and operational workbooks. Strong evidence shows structure, traceability and useful analysis.

Working spreadsheet scope

- Convert source ranges into clear tables with stable headers.
- Separate raw input, transformed data, calculation logic and presentation areas.
- Use formulas such as `SUMIFS`, `COUNTIFS`, `IF`, `XLOOKUP` or an equivalent lookup pattern where the task calls for them.
- Create pivot tables that answer defined questions at the correct level.
- Apply data types and number formats consistently.
- Flag duplicates, blanks, invalid categories and unexpected values.
- Use Power Query when repeatable import or transformation adds real value.
- Add a compact summary that links back to the underlying cells.

Workbook architecture

Sheet role	Purpose	Evidence check
Read Me	Explain the question, source, owner and refresh order.	A reviewer understands the workbook before opening calculations.
Raw Data	Preserve the original extract.	Source values remain unchanged.
Clean Data	Hold documented corrections and transformation outputs.	Each change has a rule or query step.
Analysis	Store formulas, pivots and quality checks.	Measures link to visible inputs.
Summary	Present the answer for the intended user.	The report supports a decision and avoids decorative charts.

Spreadsheet completion gate

Open the workbook on a fresh machine or account. Refresh the data connection when applicable. Check that formulas resolve, labels remain visible, charts use the intended range, and the summary matches independent totals.

EXCEL EVIDENCE RULE: A polished dashboard earns credibility when the calculation path stays visible and auditable.

SHARED FOUNDATIONS

Analyse, interpret and prepare the data

Analytical work begins with a decision question. Write that question before calculating anything. It controls which records belong, which measures matter and which limitations deserve space in the final report.

A disciplined analysis sequence

- 1 Define the unit of analysis and the period covered.
- 2 Inspect types, missingness, duplicates, category values and extreme observations.
- 3 Write each cleaning rule and preserve the original values.
- 4 Explore distributions, totals, changes and useful segment comparisons.
- 5 Select measures that answer the question at the correct grain.
- 6 Test the result against an independent total or alternate method.
- 7 Separate observed findings from interpretation and recommendation.
- 8 Record limits that could change the decision.

Statistics you should control

Know when to use counts, percentages, averages, medians, rates and change over time. Understand how an extreme value can distort an average. Distinguish a count from a rate and a percentage-point change from a percentage change. Treat correlation as an observed relationship until stronger evidence supports a causal claim.

Cleaning log

Issue	Decision to record	Validation
Missing values	Keep, exclude, replace or classify; state the reason.	Compare record counts before and after treatment.
Duplicates	Define the key and the duplicate rule.	Record removed rows and preserve an audit copy.
Invalid categories	Map source values to an approved category list.	List unmapped values for review.
Type errors	Convert dates, numbers and text with explicit rules.	Count failed conversions.
Extreme values	Investigate source and business context before treatment.	Show the effect with and without the value.

ANALYSIS STANDARD: A reviewer should be able to follow how raw records became the reported result and where judgment entered the process.

SHARED FOUNDATIONS

Report findings for a real reader

A dashboard is an interface for decisions. Every page needs an audience, a question and a reading order.

Build the report around use

- State the reporting period and refresh date.
- Define every key measure in plain language.
- Lead with the information the intended user checks first.
- Use comparisons that reveal movement, gaps or exceptions.
- Keep filters purposeful and label their effect.
- Use colour for status, selection or emphasis with consistent meaning.
- Make titles describe the finding or question.
- Place detail tables where a user needs to investigate the summary.

Power BI working scope

Import and transform data through a repeatable process. Create relationships at the correct cardinality. Build measures for calculations that must respond to filters. Keep calculated columns for row-level logic that belongs in the model. Use a date table when time analysis requires consistent periods.

Test totals under every important filter. Check blank categories, multi-select behaviour and time boundaries. A visually clean report with incorrect filter logic weakens the whole project.

Write the insight note

For each major finding, record:

- What changed or differs.
- The size and period of the effect.
- The segment or process responsible.
- The limitation that shapes confidence.
- The action or follow-up question supported by the result.

Keep the wording proportional to the evidence. A descriptive dataset can support patterns and operational recommendations. It cannot carry a causal claim without an appropriate design.

REPORTING GATE: The intended reader can identify the main result, trace its measure definition and decide what to investigate next.

CAREER LANES

Choose the work before the tool list

The retained adverts cluster around several kinds of analytical work. Job titles overlap, so choose a lane by deliverables and decision context.

Lane	Main work	Common search titles	First-build stack
Reporting and Business Intelligence	Build recurring reports, dashboards, KPI views and governed measures.	Junior Data Analyst, Reporting Analyst, BI Analyst, Junior BI Analyst, Analyst: Business Insights	SQL + Excel + Power BI
Business and Operations Insights	Analyse processes, service levels, performance drivers and business requirements.	Junior Business Analyst, Business Analyst Intern, Business Insights Analyst, Data Analyst	Excel + SQL + Power BI
Analytical Investigation	Explore patterns, segments, anomalies and outcomes; write defensible findings.	Junior Data Analyst, Data Analyst Associate, Data Analyst Intern, Marketing Analytics Analyst	SQL + Excel + statistics + Python or R
Data, AI and Automation Adjacent	Extend analysis through modelling, scripting, repeatable transformation or evaluated AI use.	Data Analyst (AI), AI Junior Analyst, Analytics and Visualisation Analyst	SQL + Python or R + BI + named data environment

The first lane carries the strongest default route because SQL and Power BI formed the leading repeated combination. The second lane fits candidates who enjoy business processes and stakeholder questions. The third lane rewards deeper investigation and written interpretation. The fourth lane belongs after the working core is secure.

Lane choice rule

Choose the lane whose weekly work you would accept, whose outputs you can build, and whose entry titles appear in locations you can target. Use the stack as a boundary for one evidence cycle.

LANE 1: REPORTING AND BUSINESS INTELLIGENCE

Build reports people can trust and reuse

This lane turns operational data into recurring visibility. The work includes extracting records, defining measures, shaping a model, building dashboard views and explaining performance changes.

Typical work

- Develop and maintain monthly or weekly reports.
- Build KPI definitions with clear business rules.
- Prepare data from spreadsheets, databases or exported systems.
- Create Power BI dashboards and supporting detail views.
- Investigate unexpected totals, missing records and reporting breaks.
- Explain changes to managers, operational teams or other report users.

Search terms

Use title variants because employers divide this work differently:

- `junior data analyst Power BI SQL`
- `reporting analyst Excel Power BI`
- `junior business intelligence analyst`
- `analyst business insights`
- `BI analyst SQL dashboard`
- `graduate data analyst reporting`

Work you should enjoy

Choose this lane when you like repeatable systems, careful metric definitions, dashboard logic and the practical detail behind a reliable monthly report. The work often rewards consistency as much as discovery.

First project direction

Build a performance-reporting product with a documented metric dictionary, SQL extraction, a clean model, an executive summary page and a detail page for investigation. Include a refresh note and validation table.

LANE 1 OUTPUT: A user can refresh the report, understand each measure, find the main exception and trace the total back to prepared data.

LANE 1: REPORTING AND BUSINESS INTELLIGENCE

Skill and evidence plan

Priority	Capability	Evidence target
Core	SQL extraction and aggregation	Saved queries that produce the reporting table and reconcile to source totals.
Core	Excel inspection and control checks	Workbook showing source checks, exception lists and independent totals.
Core	Power BI modelling and measures	Relationships, measure definitions, filter-aware calculations and a clean report.
Core	KPI communication	Metric dictionary and written summary for the intended reader.
Supporting	Power Query or repeatable transformation	Refreshable steps with renamed fields, types and quality handling.
Supporting	Tableau or Looker Studio	Comparable evidence after the primary BI workflow is complete.
Differentiator	Named database or cloud environment	Clear connection between the query source and reporting workflow.

Lane 1 completion gate

- Every KPI has a definition, grain and time basis.
- SQL outputs match an independent source check.
- Relationships and filters behave correctly.
- The dashboard separates overview from investigation.
- A refresh note explains file order and dependencies.
- The insight note identifies a decision, an exception and a follow-up check.

CV evidence pattern

Use action, analytical process, output and decision context:

`Built a Power BI service-performance report from SQL-prepared data, defined filter-aware SLA measures, reconciled dashboard totals to the source extract, and documented the weekly exception-review process.`

Keep the wording aligned with the files a reviewer can open. Replace tools or outputs with your actual implementation.

LANE 2: BUSINESS AND OPERATIONS INSIGHTS

Turn operational questions into measurable decisions

This lane sits close to teams, processes and service outcomes. The analyst clarifies a question, defines measures, inspects the workflow and presents evidence that helps someone decide what to change.

Typical work

- Translate a business problem into measurable questions.
- Define process stages, statuses, owners and service levels.
- Analyse delays, exceptions, volumes and completion patterns.
- Build Excel or Power BI reporting for operational teams.
- Document requirements, metric rules and unresolved data gaps.
- Present findings in language a non-technical stakeholder can use.

Search terms

- `junior business analyst data analytics`
- `business analyst intern Power BI Excel`
- `business insights analyst SQL`
- `graduate analyst operations reporting`
- `junior data analyst KPI reporting`
- `SAP data analytics intern`

Work you should enjoy

Choose this lane when you enjoy learning how a process runs, asking precise questions and connecting data problems to operational decisions. You should be comfortable with imperfect source data and stakeholder definitions that require clarification.

First project direction

Analyse a service workflow such as support tickets, claims, applications or requests. Define stages and SLA rules, measure where delays concentrate, build a manager view and produce an exception list for follow-up.

LANE 2 OUTPUT: A process owner can see where work accumulates, which rule defines the exception and which records require action.

LANE 2: BUSINESS AND OPERATIONS INSIGHTS

Skill and evidence plan

Priority	Capability	Evidence target
Core	Business-question framing	Short brief naming the user, decision, scope and required measures.
Core	Spreadsheet and SQL analysis	Auditable calculations, grouped results and exception records.
Core	Process and KPI definition	Status map, SLA rules, metric dictionary and assumptions.
Core	Dashboard or management report	Clear performance view with drill-through or record-level detail.
Supporting	Requirements documentation	Data-field definitions, open questions and agreed calculation rules.
Supporting	Power Query or data preparation	Repeatable treatment of dates, categories, owners and missing values.
Differentiator	Domain understanding	Evidence that explains why the metric matters in the selected process.

Lane 2 completion gate

- The process stages and measure rules are written down.
- Source fields map cleanly to each reported status.
- SLA or threshold logic handles boundaries correctly.
- Exception records can be traced to the summary.
- The report separates current performance from recommended follow-up.
- The limitations name missing fields, uncertain rules and data-quality risks.

CV evidence pattern

`Analysed a multi-stage service-request dataset in SQL and Excel, defined cycle-time and SLA rules, built a Power BI exception report, and documented the records requiring operational follow-up.`

The strongest version includes a link to the brief, query logic, report and written findings.

LANE 3: ANALYTICAL INVESTIGATION

Find patterns and explain what they mean

This lane focuses on analysis that begins with an open question. The work may involve customer behaviour, marketing performance, risk patterns, survey results or another domain where the analyst must explore before deciding what deserves attention.

Typical work

- Inspect unfamiliar datasets and define the unit of analysis.
- Write SQL queries for segments, trends, cohorts or anomalies.
- Apply descriptive statistics and careful comparisons.
- Use Python or R when the investigation benefits from scripting or repeatability.
- Test alternate explanations and document limits.
- Present findings through charts, written analysis and recommendations.

Search terms

- `junior data analyst SQL Python`
- `data analyst associate statistics`
- `data analyst intern SQL Tableau`
- `marketing analytics analyst graduate`
- `junior analyst customer insights`
- `graduate data analyst statistics`

Work you should enjoy

Choose this lane when you enjoy uncertainty, close reading of data and written reasoning. The final answer may be a qualified finding instead of a recurring dashboard. Your work should show how you moved from exploration to a defensible conclusion.

First project direction

Investigate customer, campaign or product-performance data. Define segments, compare behaviour over time, identify a material pattern and test whether the conclusion survives a reasonable alternate measure. Present the result in a short analytical report with supporting queries or code.

LANE 3 OUTPUT: A reviewer can follow the question, inspect the method, understand the finding and see the limit placed on the conclusion.

LANE 3: ANALYTICAL INVESTIGATION

Skill and evidence plan

Priority	Capability	Evidence target
Core	SQL exploration	Queries for distributions, trends, segments, cohorts or anomalies.
Core	Statistical interpretation	Correct choice and explanation of counts, rates, averages, medians and changes.
Core	Data preparation	Logged cleaning rules and validation of the analytical dataset.
Core	Written findings	Concise report connecting evidence, interpretation, limitation and action.
Supporting	Python or R	Readable script or notebook that adds depth, scale or repeatability.
Supporting	BI or visualisation tool	Charts and filters serving the investigation without hiding the method.
Differentiator	Domain framing	Metric choices and limitations grounded in the selected business context.

Lane 3 completion gate

- The question and unit of analysis are explicit.
- Data exclusions and cleaning choices are logged.
- Each reported measure matches its denominator.
- The analysis tests at least one alternate segment, period or definition.
- Visuals carry labels, units and useful comparisons.
- Findings distinguish observation, interpretation and recommendation.
- Limitations explain what the dataset cannot establish.

CV evidence pattern

Investigated customer-conversion patterns using SQL and Python, compared channel and cohort performance, validated rate denominators, and produced a written insight report with limitations and recommended follow-up analysis.

Use your actual topic and tools. Every phrase should point to visible work or a documented professional example.

DATA, AI AND AUTOMATION

Extend a secure analytics core

This specialisation appears at the edge of entry-level analytics through AI-labelled analyst roles, analytics visualisation work, scripting, data modelling and cloud exposure. It becomes useful when SQL, preparation and reporting already work together.

Suitable directions

- Automate a recurring cleaning or reporting task with Python, R or Power Query.
- Build a small data model and explain table grain, relationships and refresh logic.
- Add an evaluated text-classification or forecasting component to an ordinary analysis workflow.
- Use a named cloud or database environment and document its place in the data path.
- Compare a manual method with the automated result and record errors.
- Monitor data quality or model output with visible checks.

Evidence standard for an AI feature

State the problem, input data, expected output and evaluation method. Keep a human-reviewed sample. Record false positives, false negatives or another relevant error measure. Show how low-confidence cases are handled. Protect sensitive information and declare synthetic data when used.

An API call or generated chart carries little analytical depth by itself. Your evidence needs method, evaluation, limits and a clear decision about where automation belongs.

Entry rule

Enter this specialisation after you can query structured data, clean it, explain core measures and publish a reliable report. Keep the job search anchored to analyst titles while the specialised evidence grows.

SPECIALISATION GATE: The automated or AI-assisted step improves a defined workflow, and your evaluation shows where its output can be trusted.

LANE DECISION WORKSHEET

Commit for one evidence cycle

Selected lane:

Target role titles and search terms:

Work outputs I am choosing to build:

Stack boundary

Stack role	My choice	Evidence I will produce
Query and data access		
Spreadsheet analysis		
BI or visual communication		
Database or data environment		
Optional analytical programming		

Tools I will defer until this evidence cycle is complete:

Evidence reason

Why this lane fits the work I want to do:

Why the selected project fits the lane:

The first date when I will review this decision:

COMMITMENT: Keep this lane until the selected project reaches its completion gate and the first application review is recorded.

MILESTONE ROUTES

Foundation Builder: establish control

Milestone 1: route clarity

Objective: define the lane, target titles, stack and project boundary.

Completion evidence:

- Lane decision worksheet completed.
- At least six relevant search terms saved.
- One project blueprint selected.
- Dataset type and source rules written down.
- Weekly work blocks placed on a calendar.
- First review date recorded.

Milestone 2: SQL control

Objective: answer defined questions from structured data with auditable queries.

Completion evidence:

- Source tables and grain documented.
- Filters and calculated fields explained.
- Aggregations use the correct denominator.
- At least one join includes row-count validation.
- Nulls, duplicates and unexpected categories are checked.
- Final query output feeds the next analysis step.

Advance when

You can rebuild the query result from the raw source, explain the join and grouping logic, and reconcile the main totals independently.

Suggested pacing: use the first 20 to 30 days for route clarity, SQL control and a small working slice. Advance on evidence, even when the calendar moves faster or slower.

MILESTONE ROUTES

Foundation Builder: connect and publish

Milestone 3: spreadsheet and preparation control

Objective: turn the query output or source extract into an auditable analytical dataset.

Completion evidence:

- Raw data preserved.
- Cleaning and transformation rules logged.
- Formula, pivot or Power Query logic stays visible.
- Quality checks count missing, duplicate and invalid records.
- Summary measures reconcile to source totals.
- File structure and refresh order are documented.

Milestone 4: BI and interpretation control

Objective: build a report that answers the project question and survives filter testing.

Completion evidence:

- Data model and relationships use the correct grain.
- Measures are defined in plain language.
- Report pages follow a deliberate reading order.
- Filters and time boundaries have been tested.
- Findings identify a decision or follow-up action.
- Limitations and assumptions appear beside the result or in the project note.

Milestone 5: visible evidence

Objective: publish the completed project in a form another person can inspect.

Completion evidence:

- Accessible project link tested while signed out.
- Read Me page explains question, source, method and result.
- Queries, workbook, report files or exported views are included.
- Refresh or reproduction steps are clear.
- Screenshots or a short walkthrough show the working output.
- CV-ready project description matches the published work.

MILESTONE ROUTES

Evidence Builder: select the repair candidate

List existing projects, coursework, employment examples and dashboards. Score each item from 0 to 2 against the evidence checks.

Candidate	Accessible	Clear question	Visible process	Findings	Reproducible	Total /10

Use these anchors:

- `0`: absent, inaccessible or too vague to judge.
- `1`: present with limited context or partial support.
- `2`: clear, specific and directly inspectable.

Choose the candidate with the strongest analytical core and a repair path you can finish. A lower score can still win when the missing pieces are easy to add and the project matches your chosen lane closely.

Write the repair brief

Selected project:

Lane and target titles supported:

Highest broken evidence layer:

Files or explanations required for completion:

Deadline for the repaired public version:

MILESTONE ROUTES

Evidence Builder: repair in the right order

Repair 1: access and ownership

Publish the project through a candidate-controlled link. State your contribution when the work involved a team, employer or group assignment. Remove sensitive records and replace them with safe synthetic or anonymised data where needed.

Repair 2: question and context

Add the intended user, business question, dataset source, reporting period and definition of success. Explain any synthetic-data generation rule.

Repair 3: process and validation

Include the query logic, workbook calculations, transformation steps or script. Add counts before and after cleaning. Record checks for joins, nulls, duplicates, category mappings and metric totals.

Repair 4: findings and limitations

Write what the analysis found, why it matters, where confidence is limited and what should happen next. Make the language proportional to the data.

Repair 5: reproduction and presentation

Organise the files, add a clear Read Me, provide refresh steps, label screenshots and test the entire path from link to result.

Completion gate

- The public evidence supports the CV description.
- Another person can follow the analytical path.
- Main calculations have visible checks.
- The report answers the named question.
- Limitations remain easy to find.
- The first application date is scheduled.

MOVE FORWARD: Stop polishing when the completion gate is satisfied. Use application feedback to choose the next repair.

PROJECT BLUEPRINT 1

Performance reporting and KPI dashboard

Build this for the Reporting and Business Intelligence lane.

Scenario

A service or retail organisation needs a monthly view of volume, value, completion, exceptions and period movement. Managers need an overview plus enough detail to investigate weak results.

Dataset shape

Use transaction or service records with dates, categories, owners, locations, status and value fields. Add a separate target, category or owner table so the model requires a defensible relationship. Use public data, safely anonymised data or a declared synthetic dataset.

Build route

- 1 Define the reporting period and KPI dictionary.
- 2 Document source-table grain and relationship keys.
- 3 Prepare SQL queries for totals, trends and exceptions.
- 4 Validate source and query row counts.
- 5 Build a clean model and date logic.
- 6 Create an overview page and an investigation page.
- 7 Add refresh, validation and metric-definition notes.
- 8 Write an insight summary for the intended manager.

Required outputs

- Project brief and dataset declaration.
- SQL files in execution order.
- Validation table with independent totals.
- Power BI file or comparable report artifact.
- PDF export or screenshots for quick access.
- Metric dictionary and refresh note.
- Written findings and limitations.
- Accessible project link.

Acceptance checks

- Every reported total reconciles to a documented check.
- Filters preserve the intended denominator.
- Month comparisons use consistent periods.
- Overview and detail serve different reading needs.
- Exceptions can be traced to underlying records.
- A reviewer can reproduce the result from the included instructions.

PROJECT BLUEPRINT 2

Service operations and SLA analysis

Build this for the Business and Operations Insights lane.

Scenario

A support, maintenance or request-management team needs to understand workload, response time, resolution time, backlog and SLA breaches. The current export contains inconsistent statuses and incomplete timestamps.

Dataset shape

Use ticket or service-request records with created time, first response, resolution time, category, priority, owner, status and channel. Include an SLA rule table by priority. Declare any generated dates or synthetic records.

Build route

- 1 Map the workflow stages and valid status values.
- 2 Define business-hour or calendar-hour treatment.
- 3 Clean timestamps, priorities and category labels.
- 4 Calculate response, resolution, ageing and breach measures.
- 5 Reconcile open and closed records by period.
- 6 Build a manager summary and exception list.
- 7 Investigate the categories or owners driving delay.
- 8 Write operational actions and data-quality limits.

Required outputs

- Process map and rule definition.
- SQL or Power Query preparation steps.
- Data-quality log with unresolved records.
- Excel control workbook or validation sheet.
- Power BI operational report.
- Record-level breach or backlog export.
- Insight memo for the process owner.
- Accessible project link.

Acceptance checks

- Boundary cases follow the documented SLA rule.
- Open records calculate age consistently.
- Missing timestamps receive a visible treatment.
- Summary counts reconcile to the exception list.
- Recommendations point to a measurable process issue.
- Sensitive or synthetic data treatment is declared.

PROJECT BLUEPRINT 3

Customer or marketing performance investigation

Build this for the Analytical Investigation lane.

Scenario

A commercial team wants to understand which customer segments or marketing channels contribute to conversion, repeat activity or value. The dataset shows outcomes, while the cause remains uncertain.

Dataset shape

Use customer, campaign, session or order records with dates, channels, segments, activity and outcome fields. Include cost or exposure data when the question requires efficiency measures. Use a safe public source or declared synthetic data.

Build route

- 1 Define the unit of analysis and outcome measure.
- 2 Profile missing values, duplicates and category coverage.
- 3 Build SQL queries for trends, segments and conversion rates.
- 4 Confirm denominators and compare counts beside rates.
- 5 Use Python, R or spreadsheet analysis for deeper exploration where useful.
- 6 Test an alternate period, segment definition or outcome rule.
- 7 Build focused visuals with labelled comparisons.
- 8 Write findings, limits and the next question the team should test.

Required outputs

- Analytical question and scope note.
- Field guide and cleaning log.
- SQL queries with validation checks.
- Script, notebook or workbook used for exploration.
- Visual report or presentation-ready chart set.
- Measure definitions and denominator checks.
- Insight memo with limitations.
- Accessible project link.

Acceptance checks

- The conversion or outcome denominator is explicit.
- Small segments are identified before comparison.
- Missing or unknown categories stay visible.
- Observation and causal explanation remain separate.
- The alternate analysis tests the stability of the main finding.
- Recommendations fit the evidence and name the next validation step.

PROJECT BLUEPRINT 4

Data-quality and repeatable reporting workflow

Build this for the Data, AI and Automation Adjacent direction.

Scenario

A team receives a monthly file whose headers, categories and date formats vary. Manual preparation creates inconsistent reports. The team needs a repeatable transformation with visible quality controls.

Dataset shape

Create a set of monthly CSV or spreadsheet files with controlled inconsistencies. Include renamed columns, invalid dates, duplicate records, missing identifiers and unexpected categories. Document every synthetic variation.

Build route

- 1 Define the expected schema and accepted category values.
- 2 Preserve raw files in a dated input folder.
- 3 Build transformations in Power Query, Python or R.
- 4 Produce a quality report for failed rows and rule violations.
- 5 Load the clean output into a reporting model.
- 6 Compare refreshed totals with source control totals.
- 7 Document recovery steps when a source file fails validation.
- 8 Demonstrate a second refresh using a changed monthly file.

Optional intelligent feature

Add text classification only when the dataset contains a genuine free-text field. Create a labelled review sample, measure errors, route low-confidence cases for review and compare the result with a simple rule-based baseline.

Required outputs

- Expected-schema document.
- Raw input examples and synthetic-data declaration.
- Transformation script or query steps.
- Data-quality exception report.
- Clean output and reporting view.
- Refresh log covering at least two file cycles.
- Failure-handling and reproduction instructions.
- Accessible project link.

Acceptance checks

- Raw inputs remain unchanged.
- Every correction maps to a written rule.
- Failed records remain visible for review.
- A changed source file refreshes through the workflow.
- Control totals reconcile before reporting.
- Optional classification includes measured error and human review.

PACKAGE AND APPLY

Project completion standard

Use this gate for any DA project before placing it in an application.

Evidence area	Completion standard
Access	The submitted link opens without special permission and leads directly to the project.
Question and context	The intended user, decision, dataset and reporting period are clear.
Source and ownership	Data source, synthetic treatment and personal contribution are declared.
Preparation	Cleaning and transformation choices are visible and auditable.
Analysis	Queries, calculations and measures answer the stated question at the correct grain.
Validation	Row counts, totals, joins, filters and denominators have documented checks.
Findings	The project states meaningful results, limitations and a supported next action.
Reproduction	File order, dependencies, refresh steps and required software are explained.

Recommended project structure

- `README` with question, source, method, findings and limits.
- `data/raw` containing the untouched source or safe sample.
- `data/processed` containing the analytical output.
- `sql` containing ordered query files.
- `analysis` containing the workbook, script or notebook.
- `report` containing the BI file, export and screenshots.
- `validation` containing checks and exception outputs.
- `docs` containing the field guide, metric dictionary and refresh notes.

A platform that cannot store the complete folder can still link to the missing report artifact. Keep the main Read Me as the starting point.

RELEASE GATE: Publish when every row in the completion table has visible support and the CV description matches the files.

PACKAGE AND APPLY

Translate analytical work into CV evidence

Write project and experience bullets from four parts:

- 1 The analytical responsibility or business question.
- 2 The method and tools you used.
- 3 The output you produced and how you checked it.
- 4 The decision, user or process the work supported.

Examples

- `Built a Power BI KPI report from SQL-prepared service data, defined filter-aware measures, reconciled totals to an Excel control sheet, and produced a weekly exception view for operational review.`
- `Analysed support-request cycle times using SQL and Excel, documented SLA rules, identified the categories driving backlog, and delivered a record-level follow-up list.`
- `Investigated customer conversion by channel and cohort using SQL and Python, validated rate denominators, and wrote an insight memo covering limits and next-test recommendations.`
- `Created a repeatable Power Query workflow for inconsistent monthly files, logged failed records, reconciled refresh totals, and documented recovery steps for schema changes.`

Replace vague claims

Weak wording	Evidence-based wording
Proficient in SQL	Wrote SQL joins and grouped queries to produce a validated service-performance dataset.
Advanced Excel	Built an auditable workbook using lookup logic, pivots and source-control totals.
Power BI skills	Modelled related tables, created filter-aware measures and built overview and exception pages.
Strong analytical thinker	Compared segment rates, tested denominator changes and documented the limits on interpretation.
Excellent attention to detail	Detected duplicate records and invalid categories through logged data-quality checks.

Use metrics when you can verify them. Scope, frequency, record counts and error reductions can strengthen a bullet. Fabricated numbers damage the evidence chain.

PACKAGE AND APPLY

Build a target-role application system

Create a saved role card for the lane. It should contain:

- Primary job titles and title variants.
- Location and work-arrangement boundaries.
- Core skills that trigger a closer read.
- Supporting tools that influence priority.
- Qualification and experience wording you can target.
- Exclusion rules for senior scope, unrelated work or impossible location.
- Project link matched to the lane.
- Weekly application capacity.

Read each advert in four passes

- 1 Confirm that the duties match the analytical work you selected.
- 2 Mark required, preferred and optional skills from the actual wording.
- 3 Check qualification, experience, location and work arrangement.
- 4 Map your evidence to the central requirements and record material gaps.

Application tracker

Date	Company and title	Lane fit	Central signals	Evidence submitted	Gap observed	Status and next action

Submission check

- File names are professional and consistent.
- CV target aligns with the role.
- Project link opens from a signed-out browser.
- The strongest relevant evidence appears early.
- Claims match the submitted files.
- Contact details and dates are current.

PACKAGE AND APPLY

Turn adverts and responses into a gap review

Use a review batch of ten closely matched adverts. Count only explicit signals. Separate a recurring market gap from an unusual tool request.

Signal	Adverts naming it	Evidence level now	Decision for next cycle
SQL			
Excel or spreadsheet analysis			
Power BI or BI capability			
Analysis or statistics			
Data preparation			
Python or R			
Qualification wording			
Location or work arrangement			

Review application outcomes

Track the stage reached, including no response. After a complete batch, ask:

- Are the roles genuinely aligned with the chosen lane?
- Does the CV expose the central tools and analytical work quickly?
- Does the project link open and lead to a complete evidence trail?
- Are recurring gaps concentrated in skill, qualification, experience, location or packaging?
- Which gap can be repaired with the smallest credible evidence build?
- Which request appears too rare to influence the next cycle?

Choose one change for the next cycle. Keep the rest stable long enough to see whether the change improves the application.

REVIEW RULE: Use patterns across a targeted batch. One advert or one rejection cannot define the entire route.

YOUR 30/60/90-DAY ROUTE**Day 1 to 30: establish direction and a working slice**

- Complete the starting-point and lane worksheets.
- Save target titles, locations and role-card filters.
- Select one project blueprint and secure a safe dataset.
- Build the first SQL queries with validation checks.
- Create the raw, processed, analysis and report structure.
- Finish one end-to-end slice from source to visible result.

Day 31 to 60: complete the evidence product

- Finish preparation, analysis and quality checks.
- Build the workbook or BI report against the lane standard.
- Write findings, measure definitions and limitations.
- Organise files and add reproduction steps.
- Test the project link from a signed-out browser.
- Translate the completed work into CV evidence.

Day 61 to 90: apply, measure and improve

- Launch the target-role application routine.
- Record every advert, submission, response and recurring gap.
- Complete the first ten-advert gap review.
- Repair one material evidence weakness.
- Refresh the CV and project wording after the repair.
- Set the next review date and preserve the winning parts of the route.

Foundation Builders can use the full 90 days. Evidence Builders can compress the first period and place more time into repair, packaging and applications. Milestone completion controls progress.

PERSONAL EXECUTION PLAN

Convert the route into protected weekly work

Selected lane:

Selected project:

First visible output and due date:

Weekly block	Day and time	Task boundary	Evidence created
Build block 1			
Build block 2			
Validation block			
Packaging block			
Application block			
Review block			

This week's commitment

The exact file, query, report page or decision I will finish:

The blocker I will remove before starting:

The person, document or check that will validate the result:

First application date:

First ten-advert review date:

EXECUTION RULE: Protect the next evidence-producing block. General study fills spare capacity after the committed output is complete.

RESEARCH NOTES AND VERSION POLICY

Evidence basis

This pack uses retained South African entry-level-accessible Data Analytics adverts collected and quality-checked through 11 August 2026. Skills and tools were counted only when the source named them explicitly. Missing work-arrangement fields stayed outside the related percentage. A known geography conflict remained disclosed in the internal quality record.

The retained sample supports concept validation, route design and skill-priority decisions. Market-wide representation sits outside its scope. Individual employers will continue to use different titles, qualification rules and tool combinations.

How evidence was treated

- Repeated market signals shaped the first-build priority.
- Partial signals influenced supporting guidance.
- Rare signals remained optional or lane-specific.
- Project presentation standards came from the approved SkillSignalZA product rubric and remain subject to candidate testing.
- No skill was inferred from a related qualification, job title, library or tool.
- Customer-facing percentages use the locked internal analysis.

Version policy

SkillSignalZA should review this pack after a documented new market-data cycle, an approved rubric revision, or evidence from candidate testing that changes the usefulness of a route. The version number and evidence cutoff should move with every material update.

Final route check

- ☐ I selected one lane and saved its search terms.
- ☐ I defined a stack boundary for the current cycle.
- ☐ I chose a project and recorded its completion gate.
- ☐ I scheduled the weekly work blocks.
- ☐ I know which evidence level applies to each core skill.
- ☐ I have a tested public or submitted project path.
- ☐ I created an application tracker.
- ☐ I set the first review date.

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